

## INDUSTRIAL AUTOMATION COURSE CONTENT

Duration- 6-8 Weeks

### OBJECTIVE:

To equip learners with the theoretical knowledge and hands-on experience in automation technologies like **PLC, SCADA, HMI, sensors, drives, and robotics** widely used in Industry 4.0 environments.

### WEEK - 01

#### Introduction to Industrial Automation

- Basics of automation and its need
- Introduction to Industry 4.0
- Types of automation: Fixed, Programmable, Flexible
- Overview of industrial control systems
- Applications in automotive, manufacturing, packaging, process industries

### WEEK - 02

#### Sensors & Actuators

- Classification of sensors: proximity, temperature, level, pressure, etc.
- Analog vs digital sensors
- Actuators: pneumatic, hydraulic, electric
- Signal conditioning and interfacing
- Practice: Sensor selection for automation problems

### WEEK - 02

#### Programmable Logic Controllers (PLC)

- What is a PLC? Architecture and working
- I/O modules, power supply, communication ports
- PLC programming basics: Ladder logic, FBD, STL
- Practice: Write and simulate basic ladder logic programs

## WEEK - 04

### PLC Programming & Applications

- Timers, Counters, Latching, Interlocks
- Analog Input/Output configuration
- Real-time project: Conveyor belt, motor control, tank level system
- Hands-on with Siemens/Allen-Bradley/Mitsubishi (based on lab availability)

## WEEK - 05

### SCADA Systems

- Introduction to SCADA architecture
- HMI vs SCADA
- Tags, Trends, Alarms, Data logging
- SCADA software (e.g., Wonderware, iFIX, WinCC)
- Practice: Design a GUI for a temperature monitoring system

## WEEK - 06

### Communication Protocols & Drives

- Industrial protocols: MODBUS, PROFIBUS, Ethernet/IP
- Introduction to Variable Frequency Drives (VFD)
- Motor speed control using VFD
- PLC-Drive integration
- Practice: Configure a VFD with a PLC for motor control

## OPTIONAL

## WEEK - 07

### Industrial Robotics & IoT

- Basics of industrial robotics and pick/place automation
- 5-point and 6-point robotic arms
- Introduction to IIoT (Industrial IoT) and cloud connectivity
- Edge computing and smart factory concept

## WEEK - 08

### Mini Project & Certification

- Mini Project: Design and simulate an automated system (e.g., elevator control, bottling plant)
- Documentation: Ladder diagram, SCADA interface, sensor list
- Final Assessment & Viva
- Certification of course completion

### Tools & Technologies Covered

- **Hardware:** PLC [Siemens S7-1200, Allen-Bradley, etc.], Sensors, Relays, VFDs, HMI
- **Software:** TIA Portal, RSLogix, FactoryTalk, WinCC, iFIX, SCADA simulators
- **Protocols:** MODBUS, PROFIBUS, OPC, MQTT (for IoT)

### Course Deliverables

- Lab manuals & wiring diagrams
- PLC and SCADA program files
- Assignment sheets & quizzes
- Mini-project report template
- Certificate of Completion